

Topic 7: Modern Genetics

This topic builds on the knowledge of nucleic acids and proteins and how they are involved in gene expression. Knowledge of epigenetics, the use of stem cells and how these are contributing to medical advances are considered. The ethical implications of the use of gene technology in scientific developments are discussed.

Opportunities for developing mathematical skills within this topic include: recognising and using expressions in decimal and standard form; using an appropriate number of significant figures. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Students should:

7.1 Using gene sequencing

- i Understand what is meant by the term genome.
- ii Understand how PCR can be used to amplify DNA samples, and how these samples can be used:
 - to predict the amino acid sequence of proteins and possible links to genetically determined conditions, using gene sequencing.
 - in forensic science, to identify criminals and to test paternity, using DNA profiling.